Homework #10

This is the homework for week 11. The due date is 11:59 p.m., December 8. Late submission will automatically result in "0" grade. Please include codes for the program part in your submission.

1 Computation Questions

(**Question1):** (35 points) Suppose data $X_1, X_2, \cdots, X_n$ are i.i.d. normal $N(\mu, \sigma^2)$ random variables with μ unknown and σ^2 known. Suppose we want to test the hypothesis

$$H_0 : \mu \geq \mu_0 \text{ versus } H_1 : \mu < \mu_0,$$

with μ_0 a known number.

i. Write down the likelihood function $L(\mu)$ and log-likelihood function $\ell(\mu)$.

ii. Calculate the maximizer $\arg\max_{\mu \geq \mu_0} \ell(\mu)$.

iii. We consider the likelihood ratio test: reject H_0 if

$$\frac{\sup_{\mu \geq \mu_0} L(\mu)}{\sup_{\mu \in \mathbf{R}} L(\mu)} \leq c$$

for some constant c. Prove that this is equivalent to: reject H_0 if

$$\frac{1}{n}\sum_{i=1}^{n} X_i < C$$

for some constant C (which may not equal c).

iv. Suppose we want to control type-I error to be α, derive the constant C.

i. The likelihood function and log-likelihood function are given by

$$L(\mu) = \frac{1}{(2\pi\sigma^2)^{n/2}} \exp(-\frac{1}{2\sigma^2}\sum_{i=1}^{n}(X_i - \mu)^2), \ \ell(\mu) = -\frac{1}{2\sigma^2}\sum_{i=1}^{n}(X_i - \mu)^2 - \frac{n}{2}\log(2\pi\sigma^2).$$

ii. For

$$\ell'(\mu) = \frac{n}{\sigma^2}(\overline{X} - \mu) \Rightarrow \arg\max_{\mu \geq \mu_0} \ell(\mu) = \begin{cases} \overline{X} \text{ if } \overline{X} \geq \mu_0, \\ \mu_0 \text{ if } \overline{X} < \mu 0. \end{cases}$$

iii. Since

$$\frac{\sup_{\mu \geq \mu_0} L(\mu)}{\sup_{\mu \in \mathbf{R}} L(\mu)} = \begin{cases} 1 \text{ if } \overline{X} \geq \mu_0, \\ \exp\left(-\frac{n}{2\sigma^2}(\overline{X} - \mu_0)^2\right), \end{cases}$$

$$\frac{\sup_{\mu \geq \mu_0} L(\mu)}{\sup_{\mu \in \mathbf{R}} L(\mu)} \leq c \Rightarrow \overline{X} < \mu_0 \text{ and } \mu_0 - \overline{X} > C_1,$$

we reject H_0 if $\overline{X} < C$.

iv. Since

$$\overline{X} \sim N(\mu_0, \frac{\sigma^2}{n}), \text{ reject } H_0 \text{ if } \overline{X} < \mu_0 - \frac{\sigma}{\sqrt{n}} z_\alpha.$$

(**Question2):** (35 points) Suppose we have a set of data $Y_1, Y_2, \cdots, Y_n$, which are independent and satisfying

$$Y_i \sim N(x_i\beta_1 + \beta_0, \sigma^2), \text{ where } x_i \text{ is known.}$$

i. Write down the log $-$likelihood function $\ell(\beta_0, \beta_1)$.

ii. Derive the maximum likelihood estimator $\widehat{\beta}_0, \widehat{\beta}_1$.

iii. Derive the distribution of $\widehat{\beta}_0$ and $\widehat{\beta}_1$. Based on this, suppose we want to test the hypothesis

$$H_0 : \widehat{\beta}_0 = 0 \text{ versus } H_1 : \widehat{\beta}_0 > 0,$$

write down the formula of the P-value. Similarly, suppose we want to test the hypothesis

$$H_0 : \widehat{\beta}_1 = 0 \text{ versus } H_1 : \widehat{\beta}_1 < 0,$$

write down the formula of the P-value.

i. The log-likelihood function is

$$\ell(\beta_0, \beta_1) = -\frac{1}{2\sigma^2} \sum_{i=1}^{n} (Y_i - \beta_0 - x_i\beta_1)^2 - \frac{n}{2} \log(2\pi\sigma^2).$$

ii. Take derivative, we have

$$\frac{\partial \ell}{\partial \beta_0} = \frac{1}{\sigma^2} \sum_{i=1}^{n} (Y_i - \beta_0 - x_i\beta_1), \ \frac{\partial \ell}{\partial \beta_1} = \frac{1}{\sigma^2} \sum_{i=1}^{n} (Y_i - \beta_0 - x_i\beta_1)x_i.$$

Solve these equations, we have

$$\widehat{\beta}_1 = \frac{\sum_{i=1}^{n} (Y_i - \overline{Y})x_i}{\sum_{i=1}^{n} (x_i - \overline{x})x_i}, \text{ and } \widehat{\beta}_0 = \overline{Y} - \widehat{\beta}_1 \overline{x}.$$

iii. Form least-square estimation, we have

$$\begin{bmatrix} \widehat{\beta}_0 \\ \widehat{\beta}_1 \end{bmatrix} \sim N(\begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix}, \sigma^2 (X^\top X)^{-1}),$$

we have

$$\widehat{\beta}_0 \sim N(\beta_0, \frac{\sigma^2 \sum_{i=1}^{n} x_i^2}{n \sum_{i=1}^{n} x_i^2 - (\sum_{i=1}^{n} x_i)^2}) \text{ and } \widehat{\beta}_1 \sim N(\beta_1, \frac{\sigma^2 \times n}{n \sum_{i=1}^{n} x_i^2 - (\sum_{i=1}^{n} x_i)^2}).$$

Therefore, the P-value is

$$1-\Phi\left(\frac{\widehat{\beta}_0}{\widehat{\sigma}\sqrt{\frac{\sum_{i=1}^n x_i^2}{n\sum_{i=1}^n x_i^2-(\sum_{i=1}^n x_i)^2}}}\right)$$

for β_0 and

$$\Phi\left(\frac{\widehat{\beta}_1}{\widehat{\sigma}\sqrt{\frac{n}{n\sum_{i=1}^n x_i^2-(\sum_{i=1}^n x_i)^2}}}\right)$$

for β_1. $\widehat{\sigma}^2=\frac{1}{n}\sum_{i=1}^n(Y_i-\widehat{\beta}_0-\widehat{\beta}_1 x_i)^2$.

(**Question3):** (45 points) Suppose we have a set of observations $(x_{i1}, x_{i2}, \cdots, x_{ip}, Y_i)$, assuming that it satisfies a linear model

$$Y_i=\sum_{j=1}^p x_{ij}\beta_j+\epsilon_i \text{ where } \epsilon_i\sim N(0,\sigma^2) \text{ and } \sigma \text{ is known.}$$

In a matrix form, we have

$$\mathbf{Y}=\mathbf{X}\beta+\epsilon.$$

i. Write down the log-likelihood function, and prove the MLE for $\beta=(\beta_1,\cdots,\beta_p)^\top$ is equivalent to

$$(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{Y}.$$

ii. Suppose we want to test the hypothesis

$$H_0:\beta_1=0 \text{ versus } H_1:\beta_1>0,$$

derive the formula of the P-value for this test. Hint: $\beta_1=(1,0,\cdots,0)\beta$.

i. The log-likelihood function is given by

$$\ell(\beta_1,\cdots,\beta_p)=-\frac{1}{2\sigma^2}\sum_{i=1}^n(Y_i-\sum_{j=1}^p x_{ij}\beta_j)^2-\frac{n}{2}\log(2\pi\sigma^2).$$

Taking partial derivatives and setting them to be 0, we have

$$\frac{\partial\ell}{\partial\beta_k}=0\Rightarrow\sum_{i=1}^n(Y_i-\sum_{j=1}^p x_{ij}\beta_j)\times x_{ik}=0,$$

which implies the equation.

ii. Since

$$\widehat{\beta}_1-\beta_1\sim N(0,\sigma^2 e_1^\top(X^\top X)^{-1}e_1),$$

the P-value is given by

$$\widehat{p}=1-\Phi(\frac{\widehat{\beta}_1}{\sigma\sqrt{e_1^\top(X^\top X)^{-1}e_1}}).$$

(**Question4):** (45 points) Suppose we have observations $(x_i, Y_i), i = 1, \cdots, n$. We want to fit a quadratic curve

$$Y_i = \beta_0 + \beta_1 \times x_i + \beta_2 \times x_i^2 + \epsilon_i, \text{ where } \epsilon_i \sim N(0, \sigma^2).$$

i. Write down the log-likelihood function and derive the MLE for $\widehat{\boldsymbol{\beta}}$ (matrix notation).

ii. Assume that σ^2 is known. Derive the $1 - \alpha$ confidence interval for β_1. (Hint: Your estimator should be a linear combination of Y_i, which has normal distribution).

iii. Suppose we want to test the hypothesis

$$H_0 : \beta_1 = 0, \text{ versus } H_1 : \beta_1 > 0,$$

establish the P-value of the test.

iii. For the data (6.91, 17.52) (4.32, 22.69) (2.38, 17.61) (7.98, 14.29) (8.26, 10.77) (2.00, 12.87) (3.10, 18.63) (7.69, 16.77) (2.21, 14.97) (3.42, 19.16) (8.18, 11.15) (5.39, 22.41) (1.19, 7.50) (3.21, 19.06) (5.47, 23.89) (7.35, 16.63) (2.32, 15.09) (7.54, 14.75) (1.27, 10.75) (7.33,17.42) (8.41, 9.40) (8.72, 9.83) (6.09, 22.33) (5.30, 21.37) (7.30, 17.36), assume $\sigma^2 = 1$, derive the 95% CI for β_1. Does it cover 0? What is the P-value of the test in iii.

i. The log-likelihood function is

$$\ell(\beta_0, \beta_1, \beta_2) = -\frac{1}{2\sigma^2} \sum_{i=1}^{n} (Y_i - \beta_0 - \beta_1 x_i - \beta_2 x_i^2)^2$$

and the MLE is

$$\widehat{\beta} = (X^\top X)^{-1} X^\top Y, \text{ where } X = \begin{bmatrix} 1 & x_1 & x_1^2 \\ 1 & x_2 & x_2^2 \\ \vdots & \vdots & \vdots \\ 1 & x_n & x_n^2 \end{bmatrix}.$$

ii. For

$$\widehat{\beta}_1 - \beta_1 \sim N(0, \sigma^2 e_2^\top (X^\top X)^{-1} e_2),$$

the CI for β_1 is

$$\left[\widehat{\beta}_1 - \sigma \sqrt{e_2^\top (X^\top X)^{-1} e_2} z_{\alpha/2}, \widehat{\beta}_1 + \sigma \sqrt{e_2^\top (X^\top X)^{-1} e_2} z_{\alpha/2} \right]$$

and the P-valus is

$$1 - \Phi \left(\frac{\widehat{\beta}_1}{\sigma \sqrt{e_2^\top (X^\top X)^{-1} e_2}} \right).$$

2 Programming Questions

(40 points)

i. Generate i.i.d. data (X_i, Y_i) for $i = 1, 2, \cdots, n$, where $X_i \sim N(0, 1)$, $Y_i = X_i \times 0.5 + 0.1 + \epsilon_i$, and $\epsilon_i \sim N(0, 1)$. Estimate β_0 and β_1 using the formula in Question 2. Report the estimated β_0 and β_1.

ii. Repeat step i. for 200 times, and plot a histogram for $\widehat{\beta}_0$ and $\widehat{\beta}_1$.

iii. Suppose we want to test the hypothesis

$$H_0 : \widehat{\beta}_1 = 0 \text{ versus } H_1 : \widehat{\beta}_1 < 0.$$

Implement the procedure for calculating P-value. Repeat this step for 200 times, plot the histogram of this P-value.